

Books and References:

1. SC Gupta and VK Kapoor: Fundamentals of Applied Statistics. Sulthan Chand and sons, New Delhi.
2. Parimal Mukhopadhyay: Applied Statistics. Books and Allied (P) Ltd.
3. Box GE and Jenkins G M, Time series Analysis, Holden day

SEMESTER V

Programme	B. Sc. Statistics				
Course Code	STA5CJ301				
Course Title	Estimation				
Type of Course	Major				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary Objective	Make students to understand standard sampling distr. Calculate point estimate and its properties construction of interval estimates				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	construct various sampling distribution To understand sampling distributions and its applications	U	C	Instructor-created exams / Quiz
CO2	interpret point estimation and its properties, interval estimation The student will be able to know various methods of estimation and applying them in practical cases.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	solve parameters using various methods of estimation	Ap	P	Seminar Presentation /

				Group Tutorial Work
CO4	construct confidence intervals The students understand the concept of interval estimation and its applications	U	C	Instructor-created exams / Home Assignments
CO5	apply using software	Ap	P	One Minute Reflection Writing assignments
CO6	The student will be able to know point estimation and apply it in real life situations.	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Sampling Distributions		10	20
	1	Definitions of parameter, statistic and standard error		
	2	Exact sampling distribution Chi square distribution (derivations of distributions not required).		
	3	Mean, Variance, MGF		
	4	Mode, Additive property		
	5	Students t distribution (derivations of distributions not required)		
	6	Mean, Variance, Moments		
	7	Snedecor's F distribution: (derivations of distributions not required)		
	8	mean, variance, mode		
	9	Relationship between z, t, F and Chi square distributions.		
	10	Sampling distributions, - distribution of sample mean and variance.		
	Sections from References:			
II	Point Estimation		10	20
	11	Estimator, Estimate Properties of good Estimator Unbiasedness, Sufficiency, Consistency and Efficiency		
	12	Factorization theorem		
	13	Complete Statistic		
	14	Minimum Variance Unbiased Estimator (MVUE) Cramer-Rao inequality(statement only)		
	15	Completeness Rao-Blackwell theorem(statement only),		
	16	Lehman Scheffe theorem (statement only),		
	17	Smple problems. MVB Estimators and their applications		

	Sections from References:			
III	Methods of Estimation		20	15
	18	Method of Moments,		
	19	Method of Maximum Likelihood Estimation		
	20	Application of order statistics in estimation		
	Sections from References:			
IV	Interval Estimation		8	15
	21	Concept of Confidence Interval		
	22	Confidence Intervals for mean of Normal population Large & small sample		
	23	Confidence Intervals for Proportion		
	24	Confidence intervals for Variance of Normal population		
	25	Confidence Interval for the difference of means and proportion		
	Sections from References:			
V	Open Ended Module:		12	
	1	Understanding concepts and properties from modules 1 to 4 using softwares		
	Sections from References:			
Books and References:				
1. Goon, A.M. Gupta, M.K., and Das Gupta, B. (1980): An outline of statistical theory, Vol.I, 6th revised ed. World Press limited, Calcutta.				
2. Gupta, S.C. and Kapoor, V.K. (2014): Fundamentals of Mathematical Statistics, Sultan Chand & Sons.				
3. Rohatgi, V.K. (1984) An introduction to probability theory and mathematical statistics, Wiley Eastern.				
4. Wilks, S.S. (1962): Mathematical statistics - John Wiley & Sons.				

Programme	B. Sc. Statistics
Course Code	STA5CJ302(P)
Course Title	Sampling Methods
Type of Course	Major

Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites					
Course Summary Objective	Make students aware of statistical surveys types of sampling methods of sampling and comparing them based on efficiency of estimates				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain need and necessity of sampling	U	C	Instructor-created exams / Quiz
CO2	Distinguish between methods of sampling	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Construct sampling based on nature of population	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Examine the efficiency of estimation	U	C	Instructor-created exams / Home Assignments
CO5	Construct random samples	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45)	Marks (70)
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			+30)	
I	Statistical Surveys		10	15
	1	Census and Sample Surveys		
	2	Advantages of Sample survey over Census		
	3	Basic concepts in sampling & Types of sampling		
	4	Concepts of Probability Proportional to Size and Random Sampling		
	5	Principal steps in Sample Survey.		
	6	Sampling and non-Sampling errors		
	Sections from References:			
II	Simple random sampling		12	20
	7	Simple Random Sampling (SRS) Simple Random Sampling With Replacement. (SRSWR) Simple Random Sampling Without Replacement (SRSWOR)		
	8	Merits and demerits of Simple Random Sampling (SRS).		
	9	Methods of selecting SRS (Lottery method and Random Number method).		
	10	Estimation of Mean		
	11	Variance of estimated mean		
	12	Estimate of estimated variance.		
	13	Unbiased estimate of Population Mean Square		
	Sections from References:			
III	Stratified random sampling		15	20
	14	Need for stratification		
	15	Estimation of Mean and variance of estimated mean		
	16	Proportion and Optimum allocation:		
	17	Allocation of sample size under Proportional Allocation and variance of estimated mean		
	18	Allocation of sample size under Optimum Allocation and variance of estimated mean		
	19	Comparison of Stratified sampling over SRS		
	Sections from References:			
IV			8	15
	20	Systematic sampling:		
	21	Estimation of mean and variance.		
	22	Advantages of systematic sampling over SRS and stratified sampling.		
	23	Comparison of systematic sampling over SRS and stratified sampling.		
	24	Cluster sampling: Clusters with equal sizes		
	25	Estimation of the population mean and total,		
	26	Comparison with simple random sampling		
	Sections from References:			
V	Open Ended Module: Practical Applications		30	
	1	Selection of sample and determination of sample size. Estimation of mean and variance under SRSWR Estimation of mean and variance under SRSWOR Estimation of mean and variance under Stratified sampling using real		

		life problems.		
	Sections from References:			
Books and References:				
1. Murthy M.N (1967): Sampling theory and Methods, Statistical Publisher Society, Calcutta. 2. Des Raj (2000): sample Survey Theory, Narosa publishing house. 3. Sampath S. (2000): Sampling theory and Methods. Narosa Publishing House. 4. Sukhatme B.V (1984): Sample Survey methods and its Applications, Indian Society of Agricultural Statistics. 5. S.C Gupta and V.K Kapoor: Fundamentals of Applied Statistics. Sultan Chand & Sons.				

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